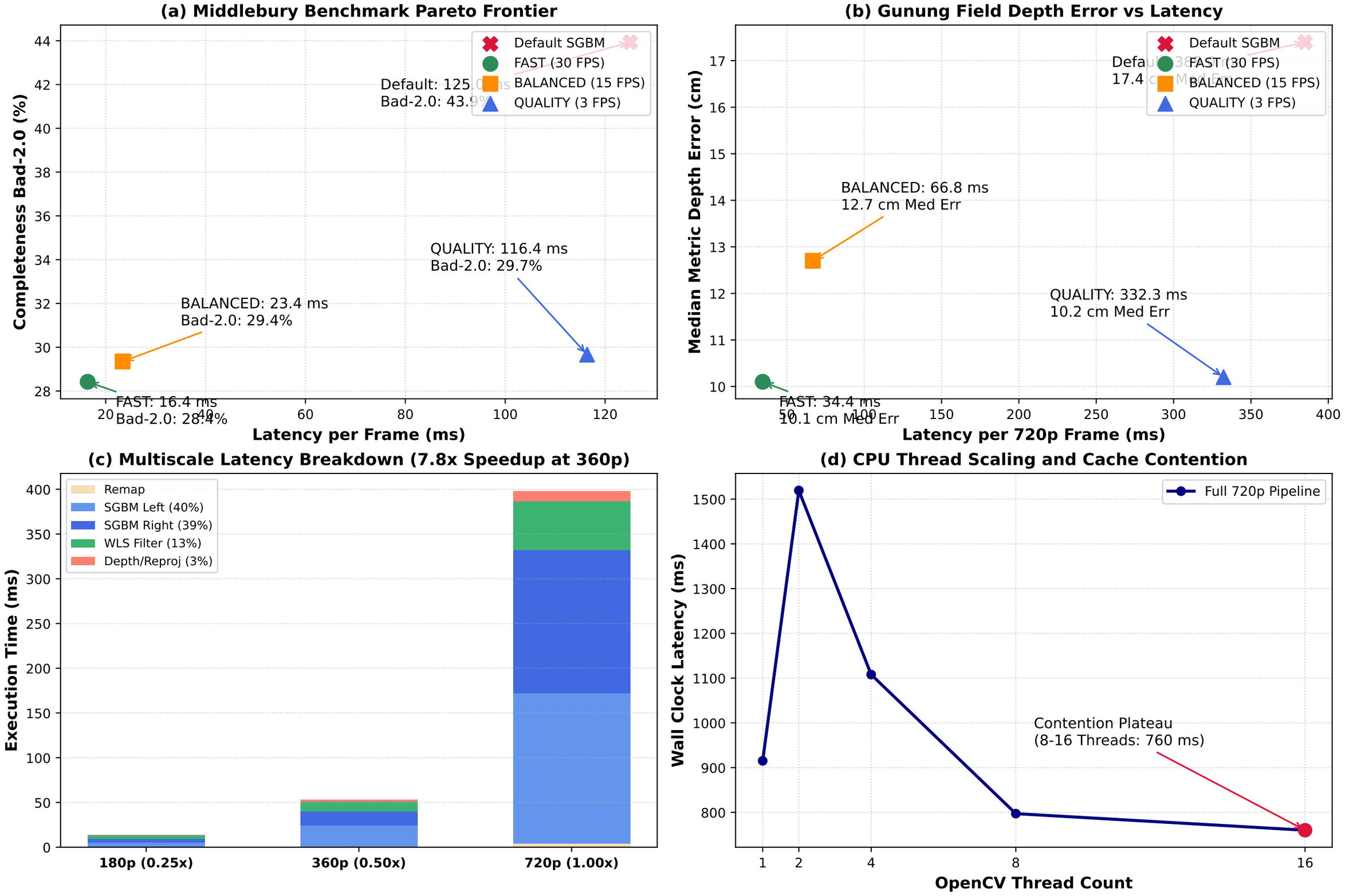
(a) Middlebury Benchmark Pareto Frontier
Completeness Bad-2.0 (%)
Latency per Frame (ms)
Default SGBM
FAST (30 FPS)
BALANCED (15 FPS)
QUALITY (3 FPS)
Default: 125.0 ms
Bad-2.0: 43.9%
BALANCED: 23.4 ms
Bad-2.0: 29.4%
QUALITY: 116.4 ms
Bad-2.0: 29.7%
FAST: 16.4 ms
Bad-2.0: 28.4%
(b) Gunung Field Depth Error vs Latency
Median Metric Depth Error (cm)
Latency per 720p Frame (ms)
BALANCED: 66.8 ms
12.7 cm Med Err
QUALITY: 332.3 ms
10.2 cm Med Err
FAST: 34.4 ms
10.1 cm Med Err
(c) Multiscale Latency Breakdown (7.8x Speedup at 360p)
Execution Time (ms)
Remap
SGBM Left (40%)
SGBM Right (39%)
WLS Filter (13%)
Depth/Reproj (3%)
180p (0.25x)
360p (0.50x)
720p (1.00x)
(d) CPU Thread Scaling and Cache Contention
Wall Clock Latency (ms)
OpenCV Thread Count
Full 720p Pipeline
Contention Plateau
(8-16 Threads: 760 ms)